Streaming intervention timing. We evaluate on the `response_loc` subset of QWENGUARDTEST (Zhao et al., 2025) (813 unsafe responses). The benchmark provides sentence-level onset spans rather than exact onset-token annotations, leaving the precise unsafe onset token undefined. Because the flagged sentence typically begins with neutral scaffolding, treating the sentence start as the unsafe onset is inaccurate. Therefore, we evaluate intervention timing against the *end* of the annotated sentence, the label-supported boundary for timely intervention.[2]

To fairly compare models with different tokenizers, we align evaluations to Qwen prefix steps by incrementally decoding Qwen token prefixes into text. Prefixes failing to decode to valid UTF-8 text are skipped. While this is slightly conservative for non-Qwen models, it ensures a standardized comparison axis. For example, let e_i denote the last token position of the annotated unsafe span, and let T_i denote the end of the response. We categorize the first trigger as **OnTime** (triggers at or before e_i), **Late** (triggers after e_i), or **Miss** (never triggers).

Latency. Additionally, we report wall-clock latency for streaming moderation. We measure average guardrail decision time and compare it to the generator's average per-token latency. Full measurement details are deferred to Appendix E.

3.2 Baselines

We compare StreamGuard against both offline and streaming guardrails. For offline post-hoc moderation, we use standard baselines such as *LlamaGuard* (Inan et al., 2023), reporting the published results. For streaming moderation, we use Qwen3Guard-Stream (Zhao et al., 2025) as a strong published boundary-detection baseline. We use its published results for the standard benchmark comparisons and the authors' officially released code for the streaming-timing and overblocking evaluations.[3]

3.3 Backbones, Transfer, and Training

We instantiate StreamGuard across multiple backbone families and sizes. Our primary scaling experiments use Llama3 (Grattafiori et al., 2024) backbones at 1B, 3B, and 8B. To study portability beyond a single tokenizer family, we additionally evaluate transfer to Gemma and Qwen backbones.

Because our supervision is prefix-based, it can be consumed either at native tokenizer positions or in text space. In the native setting, labels are attached to tokenizer-aligned prefix positions; in the transfer setting, where tokenizer coordinates differ, we reuse the same supervision by pairing decoded text prefixes with future-risk targets. We realize this with a lightweight linear head applied to the hidden state at each observed token position, which yields a risk score for the corresponding response prefix.

Training and target construction use examples from the official training splits of WILDGUARDTRAIN (Han et al., 2024), AEGIS2.0, TOXICCHAT, and BEAVERTAILS. Unless otherwise noted, all main results use a fixed reference StreamGuard target-construction setup with mean reduction, a four-model rollout pool, and a fixed supervision schedule. For the reference rollout pool, we sample continuations from Llama 3 (Grattafiori et al., 2024) 8B and 70B, and Qwen2.5 (Qwen et al., 2025) 7B and 72B, drawing four rollouts per model at temperature 0.7. We score these rollouts using Qwen3Guard-8B-strict (Zhao et al., 2025). In these datasets, rollout expansion is only necessary for unsafe prompts, as the unsafe responses present arise specifically from unsafe prompts.

Because rollout-based target construction is expensive for long responses, our reference training setup uses a budgeted supervision schedule on longer-response datasets such as WILDGUARDTRAIN, applying prefix supervision densely over an initial segment and more sparsely thereafter. We train all variants with three random seeds and report mean and standard deviation across seeds (See Appendix A for details).

[2] In 91% of examples, this is the first sentence.

[3] https://github.com/QwenLM/Qwen3Guard/commit/8a8b45a280bce0f65c322cc7086698b0d2ce4971

Model	Streaming	ToxiC	OAIMod	Aegis	Aegis2	SSTest	HarmB	WildG	Avg
PolyGuard-Qwen-7B (Kumar et al., 2025)	no	71.5	74.1	90.3	86.3	100.0	98.7	88.1	87.0
Qwen3Guard-Stream-0.6B-strict	yes	72.0	68.3	85.2	84.9	98.0	97.2	87.1	84.7
Qwen3Guard-Stream-0.6B-loose	yes	75.5	76.0	77.7	81.7	96.9	96.8	86.0	84.4
Qwen3Guard-Stream-4B-strict	yes	73.0	70.0	85.9	86.6	**99.5**	**100.0**	88.6	86.2
Qwen3Guard-Stream-4B-loose	yes	81.7	81.2	75.5	80.2	98.5	98.9	85.3	85.9
Qwen3Guard-Stream-8B-strict	yes	75.3	74.0	85.7	86.1	99.0	99.4	87.5	86.7
Qwen3Guard-Stream-8B-loose	yes	**80.1**	**80.3**	75.5	80.8	98.5	98.7	84.4	85.5
Llama3-StreamGuard-1B **(ours)**	yes	74.9± 0.3	71.5± 0.2	**89.0**± 0.3	87.7± 0.2	98.6± 0.2	94.8± 2.9	88.6± 0.2	86.4
Llama3-StreamGuard-3B **(ours)**	yes	77.7± 0.5	74.4± 0.3	87.1± 0.4	87.8± 0.1	99.3± 0.2	99.2± 0.5	89.0± 0.2	87.8
Llama3-StreamGuard-8B **(ours)**	yes	77.4± 0.5	75.0± 0.4	88.5± 0.2	**87.9**± 0.1	**99.5**± 0.0	99.7± 0.1	**89.5**± 0.2	**88.2**

Table 1: Input-moderation F1; Avg denotes the macro-average. For prior systems, we report published results. We report mean ± std over three seeds for StreamGuard models. The *Streaming* column indicates whether a model supports token-level moderation during generation. For brevity, we show only the strongest offline post-hoc model by macro F1 as a reference; full offline results appear in Appendix B.

Model	Streaming	HarmB	SRLHF	BeaverTails	XSTESTR	Aegis2	WildG	Avg
Qwen3Guard-4B-Gen-strict	no	86.7	69.8	86.6	92.7	86.1	79.5	83.6
Qwen3Guard-8B-Gen-strict	no	87.2	70.5	86.6	92.1	86.1	78.9	83.6
Qwen3Guard-Stream-0.6B-strict	yes	83.1	62.8	84.5	84.8	81.4	76.3	78.8
Qwen3Guard-Stream-0.6B-loose	yes	80.6	61.7	84.0	83.3	81.4	75.8	77.8
Qwen3Guard-Stream-4B-strict	yes	84.3	67.6	86.0	88.5	83.1	76.4	81.0
Qwen3Guard-Stream-4B-loose	yes	83.6	64.3	85.2	88.9	83.3	77.4	80.5
Qwen3Guard-Stream-8B-strict	yes	**85.0**	64.6	85.9	87.5	82.6	77.0	80.4
Qwen3Guard-Stream-8B-loose	yes	84.7	63.1	85.5	88.9	82.4	76.8	80.2
Llama3-StreamGuard-1B **(ours)**	yes	82.4± 0.2	68.0± 0.7	86.2± 0.1	84.6± 0.7	82.9± 0.3	77.2± 0.3	80.2
Llama3-StreamGuard-3B **(ours)**	yes	83.2± 0.3	68.2± 0.2	86.4± 0.0	87.2± 0.2	83.2± 0.2	**77.8**± 0.3	81.0
Llama3-StreamGuard-8B **(ours)**	yes	83.3± 0.3	**69.2**± 0.3	**86.6**± 0.2	**89.9**± 0.5	**84.5**± 0.1	**77.8**± 0.1	**81.9**

Table 2: Output-moderation F1; Avg denotes the macro-average. For prior systems, we report published results. StreamGuard models are evaluated in our simulated streaming protocol, processing responses one token at a time and counting a response as unsafe if it triggers under the debounced rule in § 3.1. We report mean ± std over three seeds for StreamGuard models. The *Streaming* column indicates whether a model supports token-level moderation during generation. For brevity, we show only the strongest offline post-hoc model by macro F1 as a reference; full offline results appear in Appendix B.

3.4 Ablation Protocol

Our ablations study practical target-construction choices while keeping the training and evaluation protocol otherwise fixed. We vary three factors: the **reduction rule** used to aggregate rollout-level safety scores into prefix targets, the **rollout source composition** used to vary the generator pool across single-model, two-model, and four-model proposal settings, and the **supervision density** used to determine how densely Monte Carlo targets are constructed along the response trajectory.

Unless otherwise noted, ablations use the Llama3-StreamGuard-8B backbone. We report the same task-appropriate metrics as in the main evaluation: F1 on moderation benchmarks, false positive rate on the benign subset of XSTEST-RESP, and OnTime / Late / Miss on QWENGUARDTEST `response_loc`.

4 Results

Input moderation. Table 1 reports input-moderation F1 across seven benchmarks. Llama3-StreamGuard-8B achieves the best average score among the compared models at 88.2, improving over Qwen3Guard-Stream-8B-strict by 1.5 F1. StreamGuard is especially strong on AEGIS2.0 and WILDGUARDTEST, while Qwen3Guard loose variants remain better on TOXICCHAT and OPENAIMODERATION, likely reflecting policy mismatch. Overall, StreamGuard remains a strong input moderator while matching or exceeding prior streaming guardrails on standard input benchmarks.

Model	QWENGUARDTEST				XSTESTR	
	F1	OnTime (%)	Late (%)	Miss (%)	F1	FPR
Qwen3Guard-Stream-0.6B-strict	95.0	88.3	2.1	9.6	83.8	7.1
Qwen3Guard-Stream-0.6B-loose	94.3	87.9	1.2	10.8	83.8	7.1
Llama3-StreamGuard-1B (**ours**)	**97.2**± 0.2	**92.9**± 0.5	1.6± 0.2	**5.5**± 0.4	**84.6**±0.7	**6.3**±0.4
Qwen3Guard-Stream-4B-strict	96.8	91.0	2.7	6.3	87.7	**4.9**
Qwen3Guard-Stream-4B-loose	94.9	88.3	2.0	9.7	**88.0**	5.4
Llama3-StreamGuard-3B (**ours**)	**97.6**± 0.0	**92.4**± 0.3	2.8± 0.2	**4.8**± 0.1	87.2±0.2	6.0±0.2
Qwen3Guard-Stream-8B-strict	95.9	89.9	2.2	7.9	88.9	4.6
Qwen3Guard-Stream-8B-loose	95.0	88.7	1.7	9.6	88.0	5.4
Llama3-StreamGuard-8B (**ours**)	**97.5**± 0.0	**92.6**± 0.2	2.5± 0.2	**4.9**± 0.0	**89.9**±0.5	**4.5**±0.3

Table 3: Streaming safety evaluation on QWENGUARDTEST and overblocking on XSTEST-RESP. Interventions are categorized as `OnTime` (at or before the unsafe sentence ends), `Late` (after the sentence ends), or `Miss` (never triggered) (§ 3.1). XSTEST-RESP reports response F1 and benign FPR. StreamGuard models report mean ± std over three seeds.

Model	X-Tok	Input Mod	Output Mod	QWENGUARDTEST				XSTESTR	
		F1 Mean	F1 Mean	F1	OnTime (%)	Late (%)	Miss (%)	F1	FPR
Qwen3Guard-Stream-0.6B-strict	No	84.7	78.8	95.0	88.3	2.1	9.6	83.8	7.1
Qwen3Guard-Stream-0.6B-loose	No	84.4	77.8	94.3	87.9	1.2	10.8	83.8	7.1
Llama3-StreamGuard-1B	No	**86.4**	80.2	97.2 ±0.2	92.9 ±0.5	1.6 ±0.2	5.5 ±0.4	84.6 ±0.7	6.3 ±0.4
Gemma3-StreamGuard-0.3B	Yes	79.9	75.3	97.6 ±0.4	91.3 ±0.4	4.0 ±0.4	4.8 ±0.8	76.1 ±2.4	11.6 ±1.2
Gemma3-StreamGuard-1B	Yes	85.1	**81.3**	**98.2** ±0.2	**92.3** ±0.3	4.2 ±0.1	**3.5** ±0.4	**87.2** ±0.0	**5.2** ±0.0
Qwen3-StreamGuard-0.6B	Yes	82.3	79.4	97.6 ±0.3	92.3 ±0.4	3.0 ±0.1	4.7 ±0.6	82.8 ±0.8	7.2 ±0.4
Qwen3-StreamGuard-1.7B	Yes	85.8	80.1	98.0 ±0.2	92.3 ±0.3	3.8 ±0.2	3.9 ±0.3	83.0 ±1.7	8.2 ±0.8

Table 4: Transfer across tokenizers and backbone families. `X-Tok` denotes prefix-level supervision transferred from a source with a different tokenizer. Input and Output Mod report macro-averaged F1 across standard suites (dataset-level breakdowns in Appendix D). QWENGUARDTEST and XSTEST-RESP protocols follow § 3.1. StreamGuard models report mean ± std over three seeds.

Output moderation. Table 2 reports response-level F1 under the streaming protocol of Section 3.1. Llama3-StreamGuard-8B achieves the best average among streaming models at 81.9 F1, outperforming Qwen3Guard-Stream-8B-strict by 1.5 points. It is strongest on SAFERLHF, XSTEST-RESP, and AEGIS2.0, while Qwen3Guard-Stream remains better on HARMBENCH. Overall, forecasting future risk yields a strong streaming output moderator and improves over prior streaming baselines while remaining competitive with offline judges that operate on complete responses.

Streaming intervention and overblocking. Table 3 reports intervention timing on QWENGUARDTEST `response_loc` and overblocking on XSTEST-RESP. On `response_loc`, all examples are unsafe, so precision is always 100 and F1 differences reflect recall. Across model sizes, StreamGuard consistently improves F1 and OnTime while reducing Miss relative to the corresponding Qwen3Guard-Stream baselines. At 8B, Llama3-StreamGuard-8B reaches 97.5 F1 and 92.6% OnTime, compared to 95.9 F1 and 89.9% for Qwen3Guard-Stream-8B-strict, while reducing Miss from 7.9% to 4.9%.

These gains do not come from indiscriminate triggering. On XSTEST-RESP, Llama3-StreamGuard-8B gives the best 8B trade-off at 89.9 F1 with 4.5 FPR, slightly improving on Qwen3Guard-Stream-8B-strict. Overall, forecasting improves timely intervention while preserving benign calibration on challenging safe responses.

Cross-tokenizer transfer. Table 4 shows that forecasting-based supervision transfers effectively across tokenizers and model families. On the output side, the strongest transferred model is Gemma3-StreamGuard-1B, which reaches 81.3 output-moderation F1, 98.2 streaming F1, and a 3.5% miss rate. Relative to the native Llama3-StreamGuard-1B baseline, it improves output-moderation average, streaming F1, miss rate, and XSTESTR F1, though its OnTime rate is slightly lower. On the input side, Qwen3-StreamGuard-1.7B is strongest

Factor	Variant	Input Mod	Output Mod	QwenGuardTest				XSTestR	
		F1 Mean	F1 Mean	F1	OnTime (%)	Late (%)	Miss (%)	F1	FPR
Reference	Mean Reduction	88.2	81.9	97.5± 0.0	92.6± 0.2	2.5± 0.2	4.9± 0.0	89.9± 0.5	4.5± 0.3
Reduction	Beta Mean	88.2	81.4	96.7 ±0.1	90.3 ±1.0	3.3 ±1.0	6.4 ±0.1	89.2 ±0.7	3.9 ±0.6
	Median	88.1	81.2	96.8 ±0.5	90.1 ±1.5	3.6 ±0.6	6.3 ±0.9	88.6 ±0.2	4.8 ±0.6
	Max	88.1	79.0	98.1 ±0.0	95.1 ±0.2	1.2 ±0.1	3.8 ±0.1	85.9 ±0.2	6.8 ±0.4
	Min	87.7	80.9	95.5 ±0.3	82.5 ±0.7	9.0 ±1.3	8.6 ±0.6	88.4 ±0.9	4.1 ±0.3
Rollout Mix	Two-model mixture (32)	88.1	81.1	96.6 ±0.1	90.3 ±0.6	3.1 ±0.4	6.5 ±0.2	89.4 ±0.0	4.3 ±0.0
	Single-model (llama3-8B)	88.2	80.9	97.6 ±0.1	93.8 ±0.3	1.6 ±0.0	4.6 ±0.3	88.5 ±2.3	5.2 ±0.8
	Single-model (qwen2.5 7B)	88.0	80.4	95.9 ±0.1	85.9 ±1.6	6.3 ±1.6	7.8 ±0.2	88.6 ±2.0	4.4 ±0.7
Sup Density	D64 S4	88.3	82.0	97.5	92.7	2.3	4.9	90.6	4.1
	D64 S8	88.5	81.9	97.6	92.9	2.5	4.7	89.7	4.9
	D32 S4	88.3	81.6	97.8	93.4	2.3	4.3	89.1	5.2
	D32 S8	88.5	82.0	97.5	92.4	2.7	4.9	90.2	4.6
	D16 S4	88.3	81.9	97.7	92.6	2.8	4.6	89.5	4.6
	D16 S8	88.4	82.0	97.7	92.5	3.1	4.4	89.5	4.6
	D1 S4	88.1	81.1	97.7	93.5	2.1	4.4	88.1	5.7
	D1 S8	88.4	81.9	97.7	93.1	2.3	4.6	89.5	4.6

Table 5: Unified ablation results for StreamGuard with the Llama-3-8B backbone. **Reduction**, **Rollout Mix**, and **Sup Density** vary the rollout-score aggregation rule, rollout-generator composition, and dense-prefix / strided-tail supervision schedule, respectively; **Reference** is the main configuration used in the paper. Prompt Mod and Resp Mod report macro-averaged F1 over the standard benchmark suites; Appendix C provides the dataset-level moderation breakdowns. QwenGuardTest and XSTest-Resp use the same streaming and overblocking protocols as in § 3.1. We report mean ± std over three seeds where shown.

among the transferred models at 85.8. Input moderation varies more across families, but the overall pattern is clear: future-risk supervision remains effective even when transferred across tokenizer families.

Wall-clock latency. Appendix E, Table 12, reports wall-clock latency for streaming moderation. Across the evaluated guardrails, decision latency ranges from 2.4 ms to 9.5 ms, corresponding to 105–420 decisions/s. Relative to Llama3-8B-Instruct, decision ratios range from 0.243 to 0.969, and relative to Llama3-70B-Instruct they range from 0.024 to 0.096. Operationally, this means that once a block decision is produced, the measured pairings stay below the regime where another token would typically be exposed to the user.

5 Ablation Studies

Table 5 studies three practical target-construction choices with the Llama3-StreamGuard-8B backbone: reduction rule, rollout-source composition, and supervision density. Overall, StreamGuard is robust to these choices. Input moderation changes little across variants, while the main trade-off is between early intervention and better benign calibration.

Reduction. Reduction has the clearest effect on operating point. Max reduction gives the strongest streaming numbers (98.1 F1, 95.1% OnTime, 3.8% Miss) but hurts overall output moderation and benign calibration, whereas min reduction is more conservative and lowers FPR at the cost of substantially worse streaming performance. Mean reduction provides the best overall balance and is therefore used as the reference setup.

Rollout mix. Rollout-source composition changes the balance between aggressiveness and robustness. Single-model Llama rollouts slightly improve streaming intervention, but mixed-generator rollouts yield better overall output moderation and calibration. This supports using a diverse rollout pool in the reference configuration.

Supervision density. Streaming performance is relatively insensitive to supervision density, suggesting that dense labels at every token are unnecessary. A modest dense prefix with a sparse tail preserves the main gains while improving efficiency, and D64 S4 gives the best XSTest-Resp trade-off in this block.